Identity, yes. Behaviour, no.

Who checks cells today, what nobody sells, and who we make stronger.

← all documents · Kat Titterton · San Francisco

Everyone adjacent certifies something about cells. Nobody certifies that cells *behave* like the field's version of the same line. Identity services prove what a line is. Instruments and AI platforms measure cells against the lab's own controls. Batch correction removes the lab-to-lab signal on purpose. Datum's product is the one thing none of them can sell: membership to a neutral, cross-lab repository of what normal looks like, refreshed with every batch a lab shares.

The map

CATEGORY	EXAMPLES	WHAT IT CERTIFIES	REFERENCE IT COMPARES TO	NEUTRAL HOLDER?
Identity and contamination testing	ATCC, core facilities, CROs	the line is what it claims; no mycoplasma	a DNA fingerprint database	no: ATCC also sells the cells
Live-cell and label-free instruments	Sartorius Incucyte	live vs dead, morphology classes	the lab's own control wells	no: sells the instrument
Phenomics and AI drug discovery	Recursion, Ardigen phenAID	a compound's effect or mechanism	batch-corrected controls, or annotated reference compounds	no: sells drug programs or services
Open consortia and methods	JUMP, batch correction	nothing; they publish data and tools	removes site and batch by design	yes, but no product and no refresh

CATEGORY	EXAMPLES	WHAT IT CERTIFIES	REFERENCE IT COMPARES TO	NEUTRAL HOLDER?
Datum	you are here	your cells still sit with the field, this quarter	every lab running the same line, conditioned on covariates	yes, by construction

The last column is the wedge. The holder of a reference must not profit from the answer.

Six questions, six tools

Slide 08, written out: each tool, the question it answers well, and what it does not see, Datum Line's own gaps included. Use all of them; the last one did not exist.

TOOL	THE QUESTION IT ANSWERS	DOES NOT SEE
DNA fingerprint (STR)	Is it the line the label says?	how the cells behave
vendor certificate	Was it fine when it shipped?	anything after shipping
plate controls	Did this plate work?	a whole lab drifting
in-house platforms	Is my own data consistent?	lab differences; it removes them
data factories	Where do I get cell data at scale?	other labs' "normal"
Datum Line	Do my cells look "normal", next to every other lab?	which cell line it is; unstained images, not yet

Phenomics platforms and batch correction remove lab differences deliberately, so biology can be compared across sites; that is correct for discovery and is exactly the coordinate the check needs to keep.

Collaborative landscape

Most of the board is a partner, not a rival: Datum Line is only valuable if the people who hold cells, images and methods feed the same reference. What each side gives and gets:

PLAYER	GIVES DATUM LINE	GETS FROM DATUM LINE	STATUS
JUMP Cell Painting Consortium and the Broad Imaging Platform	the day-one reference: eleven labs' U2OS controls, CC0	a QC use for the consortium's data; a cross-lab check the consortium set out to enable	public data in use; no agreement
Academic labs and cores	control-well images of shared lines	free membership: a check on every batch, and a baseline on day one for a new line	first ask of the pitch
Pharma and CRO screening groups	transfer runs across sites	a pass or flag before the screen, not four months into it	first paying members
Cell banks and vendors	lot and passage metadata	early warning when a lot drifts across labs; a neutral stamp to ship with a lot	channel, later
STR providers	identity result	a behaviour line beside identity on one report	complement
Instrument makers	image formats, acquisition metadata	a reason to keep brightfield control wells; training data from the red-teamed reference	channel, later
ML groups (batch correction, foundation models)	methods	a benchmark that grades methods on keeping the lab coordinate and catching planted failures	collaborators wanted

None of these are partners yet. The deck names coauthor institutions only ("Published with"), never as partners.

Identity and contamination testing

This is the closest existing habit, and it answers a different question. ATCC's human STR service fingerprints a line's DNA and compares it with a reference profile; it also sells PCR mycoplasma testing [1][2]. ATCC services · STR method

- **What it proves:** the cells are the line on the label, and they are clean.
- **What it can't see:** two correctly identified, mycoplasma-free cultures of the same line that behave differently. ATCC's own guidance cites low- and high-passage LNCaP cells responding divergently to the same hormones, which is exactly the gap. ATCC
- **Habit worth borrowing:** ATCC tells labs to re-test on receipt, after ten passages, after banking and "when in doubt". Labs already accept a recurring check. ATCC blog
- **Relationship to Datum:** complement, not competitor. Identity first, behaviour second. Pitch Datum as the second line on the authentication record a lab already files.
- **Neutrality:** ATCC distributes the cells it authenticates, which is fine for identity and a conflict for behaviour.

Live-cell and label-free imaging instruments

Instrument makers already sell label-free AI. Sartorius, the clearest example checked here, references the lab's own wells. Sartorius's Incucyte AI Cell Health module classifies live and dead cells from unlabelled images with a pretrained network. Its Advanced Label-Free Classification add-on trains a classifier on the plate's own live and dead control wells. The 2024B release added normalising to specific wells [5][6][7]. AI Cell Health · Label-free classification · v2024B

- **What they prove:** what happened on this plate, relative to this plate.
- **What they can't see:** board A1's bug. If the control wells drifted, the classifier learned the drift.
- **Threat:** highest capability to copy the analytics. Lowest credibility as the holder, because the answer "your cells drifted" can land on their own instrument.
- **Relationship to Datum:** channel partner. Brightfield and phase images from installed instruments are the cheapest possible input to the check. The long game is to be the reference every instrument vendor reads from.

Phenomics and AI drug discovery platforms

These companies have the most images and the best models, and they treat lab-to-lab variation as noise to remove. Recursion trains the Phenom models on its own microscopy and released OpenPhenom-S/16, trained on RxRx3 and JUMP. The model card tells users to batch-correct embeddings with TVN, fit on control images across batches, before the features work [8][9]. Recursion release · OpenPhenom model card

Ardigen's phenAID compares client Cell Painting data with annotated reference profiles to predict activity and mechanism, and hosts a JUMP data explorer [10][11]. phenAID · JUMP explorer

- **What they prove:** what a compound does, after the lab signal is scrubbed out.
- **What they can't see:** the scrubbed signal. Appendix A3 shows that centring a plate on its own controls drops lab identity from 99.9% to 24%; that is the coordinate the check needs.
- **Threat:** Recursion could publish a reference. It would be tied to its own lines, conditions and incentives.
- **Relationship to Datum:** members and data partners. Their models get better inputs when the plates feeding them are aligned across labs, read by API.
- **Adjacent, not listed as competitors:** Axiom Bio builds liver-toxicity prediction on primary hepatocyte imaging. biopharmatrend It is a drug-safety product, not a cell-state reference.

Open consortia, public data and batch-correction methods

The open world built the evidence and then, by design, threw away the product. JUMP brought ten pharma companies, six technology companies and two non-profits together to image U2OS cells under one protocol [13]. JUMP A published benchmark of batch correction on that data tested ten methods and found Harmony and Seurat RPCA best at removing batch effects [14]. Arevalo 2024 In ML, WILDS includes cross-experiment cell imaging (RxRx1) among its distribution-shift benchmarks [15]. WILDS

- **What they prove:** the effect is real and large.
- **What they don't do:** run a service, refresh a reference, or stand behind an answer for a specific lab.
- **Relationship to Datum:** foundation and first allies. Build on JUMP, cite it, and offer the consortium the repository as its natural next step: new plates from its members, checked against the map it built.

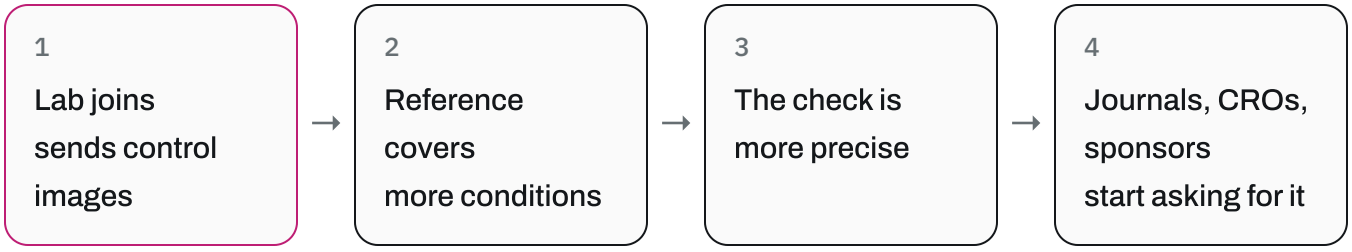
Standards bodies and reference materials

The standards world already made identity a norm, and it did so with a neutral partner. ATCC developed its mouse STR profiling with NIST and tested it across ten independent labs [4]. ATCC standards The US cost of irreproducible preclinical work was estimated at about US$28B a year, with reagents and reference materials, including misidentified lines, the largest category [17]. PLoS Biology · Science News

- **Relationship to Datum:** the path to a behavioural standard runs through the same kind of body. Plan for a standards partner, not a standards fight.
- **Open question:** whether a phenotypic reference material, a physical control line run by every lab, belongs in the product.

Why neutrality is the moat, and who moves next

The model is not the moat; the reference is, and only a neutral party can hold it. Your own co-authored benchmark work shows simple cell counts match rich profiles on many standard tasks, so feature extraction will commoditise [16]. Nature Communications 2026 A reference compounds instead: each lab that joins makes every other lab's estimate tighter.



The loop only turns if nobody fears the holder. That rules out anyone who sells cells, media, instruments or drug programs.

WHO COULD MOVE	LIKELY MOVE	DATUM'S ANSWER
Sartorius or another instrument vendor	in-instrument "field comparison"	limited to its own installed base; can't certify its own drift
Recursion	publish a reference from RxRx data	its lines and conditions; a drug company grading everyone else

WHO COULD MOVE	LIKELY MOVE	DATUM'S ANSWER
ATCC	add phenotype to authentication	sells the cells; natural partner for identity plus behaviour
JUMP or the Broad	publish controls as an atlas	welcome: an atlas is not a refreshed, per-lab service; partner and build on it

Short-term loss that buys a structural win: give the reference's summary statistics away to the open community, and sell membership: the check on every batch and the aligned images.

The market, bottom up

No public count exists of organisations running high-content imaging on cell lines, so the model below starts from assumptions and says so. What the public record gives: the high-content screening market is reported at about US$1.8B in 2025 by one analyst and US$3.4B in 2024 by another, with instruments the largest segment [L1, L2]; the JUMP consortium that produced the public map was ten pharmaceutical companies, the Broad Institute and Ksilink, with six supporting technology companies [L3]; and a 2024 systematic review traces a decade of Cell Painting adoption across academia and industry without a count of sites [L4]. At a few hundred thousand dollars per system and a five-to-seven-year life, the instrument revenue alone implies an installed base in the thousands of systems. The first row below is therefore conservative: it counts organisations that screen continuously on cell lines, not every core that owns an instrument. Labs come in two kinds: some run a few lines deep (a primary line and a co-culture, say), some run panels of hundreds to thousands of genotypes wide. A price per line fits the first; a site licence fits the second; both are rows. Cores and academic labs are free in this model; they are the map, not the revenue.

ASSUMPTION	LOW	BASE	HIGH
organisations screening a few lines deep	200	400	800
lines each runs per quarter	2	4	8
price per line per quarter, $k	5	10	15
organisations screening panels of hundreds to thousands of lines	10	25	50
site licence per year, $k	50	150	300

ASSUMPTION	LOW	BASE	HIGH
companies buying cell data they did not make	50	150	300
purchased datasets each places on the map per year	2	4	8
price per dataset membership, $k	10	25	50
benchmark partners	4	10	20
membership per partner per year, $k	50	100	200
PER-LINE CHECKS AND SITE LICENCES, PER YEAR	$8.5M	$67.8M	$399M
DATASET MEMBERSHIPS ON PURCHASED DATA, PER YEAR	$1M	$15M	$120M
BENCHMARK MEMBERSHIP, PER YEAR	$200K	$1M	$4M
SERVICEABLE, PER YEAR, AT FULL ADOPTION	$9.7M	$83.8M	$523M

A tenth of the base case is about $8.4M a year, the five-year bar. The anchor on the value side is one screen run on cells that had moved: about US$2.6M of work to redo. Prices are the proposal in the business model section; the two rows that matter most, organisations and lines per organisation, are replaced by counts from the first three partner labs and from the instrument installed base as soon as either exists. The same table is board A22 in the deck.

What the plan costs

Eighteen months, planned line by line and asked at 2×, on the rule that most things take two to three times the time and money they should. Planned $0.9M; asked about $1.8M. No lab of our own: partner labs image the controls they already take, and the benchmark runs where the cells already are or at a CRO. The benchmark playground is inside the plan as milestone three, not a later round. Board A21 in the deck carries the same lines.

LINE	WHAT	PLANNED
leader, to run it	18 months	$180k
scientific lead	18 months	$180k

LINE	WHAT	PLANNED
ML engineer	from month 7	$160k
payroll overhead	20% of salaries	$104k
benchmark playground	~60 plates · planted faults	$120k
partner-lab bench time	3 labs · controls they already image · reagent top-ups · shipping	$30k
compute and storage	images · models · hosting · API	$45k
legal, IP, insurance, accounting	incorporation to first contract	$45k
travel and partner visits	2 benchmark partners · 3 labs	$20k
AS PLANNED		$0.88M
ASKED, AT 2×		$1.77M

Partner labs

No partner lab is named until an agreement is public. U2OS is the on-ramp to the public map; a lab on any other line starts that line's map from its first batch, and it becomes a cross-lab map when a second lab on the same line joins.

Risks and answers

RISK	WHY IT'S REAL	ANSWER
Cold start: no reference without labs, no labs without a reference	network products stall at zero	seed from JUMP's public controls; first members work on lines JUMP already covers (U2OS)

RISK	WHY IT'S REAL	ANSWER
The signal is the instrument, not the cells	partly: every feature family names the lab, but labs on one microscope model are still told apart (99.99%)	the same line run on many instruments is precisely the design that separates them; make it experiment one
Label-free doesn't carry the signal	all evidence so far is fluorescence	test brightfield on plates that already carry both; say it's a bet until then
Shelf life is thin	one lab, 219 plates, and drift still rising at four months	re-measure at every partner; the refresh interval becomes a published number
Labs won't share	data fear	only DMSO control images are shared; no compounds, no results
A big vendor copies it	they have images and distribution	they can't be neutral about their own drift; partner as a data channel instead
Nobody pays until someone requires it	QC is a cost centre	sell first where results must transfer: CROs, multi-site programs, tech transfer

Sources

Checked through web search on 22 Sept 2026; vendor claims are the vendors' own pages.

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3. ATCC: know your cells
4. ATCC standards
5. Sartorius Incucyte AI Cell Health
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9. OpenPhenom model card
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14. Arevalo et al. 2024, batch correction
15. Koh et al. 2021, WILDS
16. Seal et al. 2026, counting cells
17. Freedman et al. 2015, economics of reproducibility
18. High Content Screening Market Size, Share and Forecast 2034. Credence Research. credenceresearch.com/report/high-content-screenings-market
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